

SUMS Proofs Workshop: Winter 2026 Solutions

Note: Problems marked with (*) are difficult. (**) indicates *very difficult*.

Problems from Fall's Workshop

PROBLEM 1 Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for every $n \in \mathbb{Z}^+$.

Proof. Base case $n = 1$: $1^2 = 1 = \frac{1 \cdot 2 \cdot 3}{6}$. Assume true for $n - 1$. Then

$$\sum_{k=1}^n k^2 = \frac{(n-1)n(2n-1)}{6} + n^2 = \frac{2n^3 + 3n^2 + n}{6} = \frac{n(n+1)(2n+1)}{6},$$

completing the induction. □

PROBLEM 2 Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$.

Proof. There are two cases.

Case 1: $x \leq y$. Then $\max(x, y) = y$ and $\min(x, y) = x$, so

$$\max(x, y) + \min(x, y) = y + x = x + y.$$

Case 2: $y < x$. Then $\max(x, y) = x$ and $\min(x, y) = y$, so

$$\max(x, y) + \min(x, y) = x + y.$$

In both cases the sum equals $x + y$, establishing the identity. □

PROBLEM 3 (*) Assume that the following statement is true: if Micah dislikes pineapple pizza then he likes pineapple pizza. Does he like pineapple pizza?

Proof. Let

p : Micah dislikes pineapple pizza

The hypothesis is $p \Rightarrow \neg p$. Then we can construct the following truth table:

p	$\neg p$	$p \Rightarrow \neg p$
T	T	T
F	F	T
F	T	T

Constructing the truth table shows the only way $p \Rightarrow \neg p$ can hold is when p is false (otherwise p and $\neg p$ would both be true or both false, impossible). Thus p is false, so $\neg p$ is true, i.e. Micah likes pineapple pizza. □

PROBLEM 4 (**) The *Fibonacci sequence* is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}$$

for $n \geq 2$. Show that every positive integer can be written as the sum of distinct Fibonacci numbers. Distinct means the sum cannot have repeated values; For example, $2 = 1 + 1$ is not a valid sum even though $F_1 = 1$ and $F_2 = 1$. But $2 = 2 = F_3$ is.

Proof. We proceed by strong induction. The base cases $1 = F_1$ and $2 = F_3$ hold. Suppose for induction that all positive integers less than $n > 2$ can be written as the sum of distinct Fibonacci numbers.

We show this for n . Let F_k be the largest Fibonacci number with $F_k \leq n$. Then $F_{k+1} > n \geq F_k$. Letting $r := n - F_k$,

$$0 \leq r < F_{k+1} - F_k = F_{k-1} < n.$$

If $r = 0$, $n = F_k$ and we are done. Else, $r < n$ so by the inductive hypothesis, r can be expressed as the sum of distinct Fibonacci numbers, none of which can be F_k as $r < F_{k-1}$. Thus $n = F_k + r$ shows n can be written as the sum of distinct Fibonacci numbers.

Thus by induction the statement holds for $n \in \mathbb{N} \setminus \{0\}$. □

PROBLEM 5 (**) What is the minimum number of breaks to break up a chocolate bar made of $m \times n$ little squares into those mn little squares?

Proof. Every time you break a chocolate bar, you increase the number of pieces by 1. Then you need $mn - 1$ breaks no matter what. □

Starter Problems

PROBLEM 6 Consider the following statement: For all positive integers x , $x^2 + x + 41$ is prime. Is the statement true or false? Prove it.

Proof. It is false. The counterexample $x = 41$ gives $x^2 + x + 41 = 41(41 + 1 + 1)$ which is clearly not prime. □

PROBLEM 7 Prove that no integer n , when squared, ends in 2, 3, 7, or 8.

Proof. The last digit of n^2 depends only on the last digit of n . Checking 0 through 9 gives squares ending in

$$0, 1, 4, 9, 6, 5, 6, 9, 4, 1.$$

Thus a square can only end in 0, 1, 4, 5, 6, or 9, never in 2, 3, 7, or 8. □

PROBLEM 8 Let $n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$. Prove that $n!$ is divisible by $2^{n/2}$ for all even positive integers n .

Proof. If n is even, then among the factors $1, 2, \dots, n$ there are exactly $n/2$ even numbers. Each even factor contributes at least one factor of 2, so the product $n!$ has at least $n/2$ factors of 2. Hence $2^{n/2}$ divides $n!$. $\square$

PROBLEM 9 Prove that if a divides b and b divides a , then $|a| = |b|$. Is it true that $a = b$ too? Why or why not?

Proof. If $a \mid b$ and $b \mid a$, then there are integers m, n with $b = am$ and $a = bn$. Substituting gives $a = amn$, so $a(1 - mn) = 0$. If $a = 0$, then $b = 0$ and $|a| = |b|$. If $a \neq 0$, then $mn = 1$, so m and n are 1 or -1 . Hence $b = \pm a$ and $|a| = |b|$. It is not always true that $a = b$; for example $a = 2$ and $b = -2$. $\square$

PROBLEM 10 Prove that if $A \subseteq B$ then $B^c \subseteq A^c$, where A^c is the complement of A .

Proof. Let $x \in B^c$. Then $x \notin B$. Since $A \subseteq B$, $x \notin A$. Hence $x \in A^c$, so $B^c \subseteq A^c$. $\square$

PROBLEM 11 Show that in any group of n people, where each person is friends with some of the other $n - 1$, there are always two people with the same number of friends.

Proof. Each person has between 0 and $n - 1$ friends. It is impossible to have someone with 0 friends and someone with $n - 1$ friends at the same time, since if a person is friends with everyone then no one has 0 friends. Thus there are at most $n - 1$ possible friend counts among n people. By the pigeonhole principle, two people share the same number of friends. $\square$

PROBLEM 12 You have a pile of 100 coins. Every move, you may replace a pile of size n with two piles of a and b coins, where $a + b = n$ and $a \neq b$. Can you end with 100 piles of 1 coin? Prove or disprove.

Proof. We show this is impossible for any $n > 1$ by strong induction on n . For $n = 2$, the only split is $1 + 1$, which is not allowed. Assume no pile of size m with $2 \leq m < n$ can be split into all 1's using only unequal splits. For a pile of size n , any allowed split gives $a + b = n$ with $a \neq b$ and $1 \leq a, b < n$. If $a = 1$ then $b = n - 1 > 1$, and if $a > 1$ then both parts exceed 1. In all cases at least one part is a pile of size m with $2 \leq m < n$, which cannot be fully split into 1's by the induction hypothesis. Hence no pile of size $n > 1$ can be reduced to all 1's. In particular, starting from 100 it is impossible. $\square$

PROBLEM 13 Let $\gcd(a, b)$ denote the greatest common divisor between integers a and b . Prove that $\gcd(a, b) = \gcd(a, b - a)$ for all integers a and b .

Proof. Let $d = \gcd(a, b)$. Then $d \mid a$ and $d \mid b$, so $d \mid (b - a)$. Hence d is a common divisor of a and $b - a$. Conversely, if c divides both a and $b - a$, then c divides $(b - a) + a = b$. Thus the common divisors of (a, b) and of $(a, b - a)$ are the same, so their greatest common divisors are equal. $\square$

PROBLEM 14 Show that $2 + 4 + 6 + \dots + 2n = n(n + 1)$. Can you do it without induction?

Proof.

$$2 + 4 + \cdots + 2n = 2(1 + 2 + \cdots + n) = 2 \cdot \frac{n(n+1)}{2} = n(n+1).$$

□

PROBLEM 15 Prove that if $A \times B = A \times C$ and $A \neq \emptyset$, then $B = C$.

Proof. Choose $a \in A$. If $b \in B$, then $(a, b) \in A \times B = A \times C$, so $b \in C$. Thus $B \subseteq C$. The same argument with C and B swapped shows $C \subseteq B$, so $B = C$. □

Harder Problems

PROBLEM 16 (*) Prove that if $2^n - 1$ is prime, then n is prime. Hint:

$$2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 1).$$

Proof. We prove the contrapositive. If n is composite, write $n = ab$ with $a, b > 1$. Then

$$2^n - 1 = 2^{ab} - 1 = (2^a)^b - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 1),$$

so $2^a - 1$ is a nontrivial divisor of $2^n - 1$. Hence $2^n - 1$ is not prime. Therefore, if $2^n - 1$ is prime, then n is prime. □

PROBLEM 17 (*) Prove that if $a \equiv b \pmod{m}$ then $a^k \equiv b^k \pmod{m}$ for all positive integer k .

Proof. We use induction on k . For $k = 1$, $a \equiv b \pmod{m}$ is given. Assume $a^k \equiv b^k \pmod{m}$. Then

$$a^{k+1} - b^{k+1} = a \cdot a^k - b \cdot b^k = a(a^k - b^k) + (a - b)b^k.$$

Both terms are divisible by m , since $m \mid (a^k - b^k)$ and $m \mid (a - b)$. Hence $m \mid (a^{k+1} - b^{k+1})$, so $a^{k+1} \equiv b^{k+1} \pmod{m}$. By induction, the statement holds for all k . □

PROBLEM 18 (*) Prove $\sqrt{6}$ is irrational.

Proof. Suppose $\sqrt{6} = \frac{a}{b}$ with integers a, b in lowest terms and $b \neq 0$. Then $6b^2 = a^2$, so $2 \mid a$ and $3 \mid a$, hence $6 \mid a$. Write $a = 6k$. Then $6b^2 = 36k^2$, so $b^2 = 6k^2$, which implies $6 \mid b$. This contradicts that a/b is in lowest terms. Therefore $\sqrt{6}$ is irrational. □

PROBLEM 19 (*) Show that every squared integer can be written as the sum of consecutive odd numbers.

Proof. We claim $n^2 = 1 + 3 + 5 + \cdots + (2n - 1)$. For $n = 1$ this is true. If it holds for n , then

$$(n+1)^2 = n^2 + 2n + 1 = (1 + 3 + \cdots + (2n - 1)) + (2n + 1),$$

so it holds for $n + 1$. Thus every square is a sum of consecutive odd numbers. □

PROBLEM 20 (*) Start with the number 0. Every move, you can add 3 or 5 to it. Show that, given enough moves, you can make every integer ≥ 8 , starting from 0 every time.

Proof. We can reach $8 = 3 + 5$, $9 = 3 + 3 + 3$, and $10 = 5 + 5$. For any integer $n \geq 8$, if we can reach n , then we can reach $n + 3$ by adding 3. Hence all integers $n \geq 8$ are reachable. $\square$

PROBLEM 21 (*) Prove that every positive number can be written as the sum of distinct powers of 2 (including $2^0 = 1$).

Proof. We use strong induction on $n \geq 1$. The statement holds for $n = 1 = 2^0$. Assume it holds for all integers less than n . Let $2^k \leq n < 2^{k+1}$. Then $n = 2^k + r$ with $0 \leq r < 2^k$. If $r = 0$ we are done. Otherwise, by the induction hypothesis r is a sum of distinct powers of 2, none of which equals 2^k since $r < 2^k$. Thus n is a sum of distinct powers of 2. $\square$

PROBLEM 22 (*) Ten lamps are arranged in a line, all turned off. You may pick a lamp to toggle between on and off, but it will also toggle its immediate neighbors. Is it possible to only have the 5th lamp from the left on? Prove or disprove.

Proof. Let $x_i \in \{0, 1\}$ indicate whether we toggle lamp i an odd number of times. The final state of lamp i is $x_{i-1} + x_i + x_{i+1} \pmod 2$, with missing neighbors omitted. We want only lamp 5 on, so we solve

$$\begin{aligned}x_1 + x_2 &= 0 \\x_1 + x_2 + x_3 &= 0 \\x_2 + x_3 + x_4 &= 0 \\x_3 + x_4 + x_5 &= 0 \\x_4 + x_5 + x_6 &= 1 \\x_5 + x_6 + x_7 &= 0 \\x_6 + x_7 + x_8 &= 0 \\x_7 + x_8 + x_9 &= 0 \\x_8 + x_9 + x_{10} &= 0 \\x_9 + x_{10} &= 0.\end{aligned}$$

Solving in $\mathbb{Z}_2$ gives $x_6 = x_7 = x_9 = x_{10} = 1$ and all other $x_i = 0$. Thus toggling lamps 6, 7, 9, 10 once each produces exactly lamp 5 on. So it is possible. $\square$

PROBLEM 23 (*) Prove that $n^5 - n$ is divisible by 30 for every positive integer n . Hint: You may use Fermat's little theorem: If p is prime, for any integer a we have $p | a^p - a$.

Proof. We show divisibility by 2, 3, and 5. If n is even, then $n^5 - n$ is even. If n is odd, then $n^5 - n = n(n^4 - 1)$ is even since $n^4 - 1$ is even. So 2 divides $n^5 - n$. Modulo 3, we have $n^3 \equiv n$, so $n^5 = n^3 n^2 \equiv n n^2 = n^3 \equiv n$, hence 3 divides $n^5 - n$. Modulo 5, by Fermat's little theorem $n^5 \equiv n$, so 5 divides $n^5 - n$. Since 2, 3, 5 are pairwise coprime, 30 divides $n^5 - n$. $\square$

PROBLEM 24 (*) Prove that, among any $n + 1$ integers, not necessarily consecutive, at least two have a difference divisible by n .

Proof. Consider the n possible remainders modulo n . Among $n + 1$ integers, two share the same remainder. Their difference is divisible by n . $\square$

PROBLEM 25 (*) Show that $1 + 3 + 9 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$ without induction.

Proof. Let $S = 1 + 3 + 3^2 + \dots + 3^{n-1}$. Then $3S = 3 + 3^2 + \dots + 3^n$. Subtracting gives $3S - S = 3^n - 1$, so $S = \frac{3^n - 1}{2}$. $\square$

Challenging Problems

PROBLEM 26 (**) Prove that any integer sequence of length n contains a subsequence (i.e. a consecutive subset) with sum divisible by n .

Proof. Let the sequence be $a_1, \dots, a_n$ and define partial sums $s_k = a_1 + \dots + a_k$ for $1 \leq k \leq n$. If any s_k is divisible by n , the subsequence $a_1 + \dots + a_k$ works. Otherwise, the n numbers $s_1, \dots, s_n$ have remainders $1, \dots, n - 1$ modulo n , so two must share a remainder. Then $s_j - s_i$ is divisible by n for some $i < j$, and the subsequence $a_{i+1} + \dots + a_j$ has sum divisible by n . $\square$

PROBLEM 27 (**) Prove that the decimal expansion of a rational number will eventually repeat.

Proof. Let $x = \frac{p}{q}$ with integers p and $q > 0$. Long division produces remainders r_k with $0 \leq r_k < q$, and each next digit is determined by $10r_k/q$. There are only q possible remainders, so some remainder repeats. From the first repetition onward, the digits repeat, giving an eventually repeating decimal expansion. $\square$

PROBLEM 28 (**) Prove that there is no smallest positive rational number.

Proof. If $r > 0$ is rational, then $\frac{r}{2}$ is also rational and satisfies $0 < \frac{r}{2} < r$. Thus no positive rational can be the smallest. $\square$

PROBLEM 29 (**) Prove that for all real numbers $x_1, \dots, x_n$, and for all positive integers n , then

$$(x_1 + \dots + x_n)^2 \leq n(x_1^2 + \dots + x_n^2).$$

Proof. We use the inequality

$$0 \leq \sum_{i < j} (x_i - x_j)^2.$$

Expanding gives

$$0 \leq (n-1) \sum_{i=1}^n x_i^2 - 2 \sum_{i < j} x_i x_j.$$

Thus

$$\left(\sum_{i=1}^n x_i \right)^2 = \sum_{i=1}^n x_i^2 + 2 \sum_{i < j} x_i x_j \leq n \sum_{i=1}^n x_i^2,$$

which is the desired inequality. $\square$

PROBLEM 30 (**) Let $\alpha \in \mathbb{R}$ such that $\alpha + \frac{1}{\alpha} \in \mathbb{Z}$. Prove that $\alpha^n + \frac{1}{\alpha^n} \in \mathbb{Z}$ for any $n \in \mathbb{N} \cup \{0\}$.

Proof. Let $t = \alpha + \frac{1}{\alpha} \in \mathbb{Z}$ and define $s_n = \alpha^n + \frac{1}{\alpha^n}$. Then $s_0 = 2$ and $s_1 = t$. Also,

$$s_{n+1} = \alpha^{n+1} + \frac{1}{\alpha^{n+1}} = t \left(\alpha^n + \frac{1}{\alpha^n} \right) - \left(\alpha^{n-1} + \frac{1}{\alpha^{n-1}} \right) = ts_n - s_{n-1}.$$

By induction on n , $s_n \in \mathbb{Z}$ for all $n \in \mathbb{N} \cup \{0\}$. □

Some AM-GM Identities

Please do the problems in this section in order.

PROBLEM 31 (**) This is known as the inequality of arithmetic and geometric means (AM-GM). For all positive real numbers a and b ,

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Hint: $(\sqrt{a} - \sqrt{b})^2$.

Proof. Since $a, b > 0$, we have $(\sqrt{a} - \sqrt{b})^2 \geq 0$. Expanding gives $a + b - 2\sqrt{ab} \geq 0$, so $\frac{a+b}{2} \geq \sqrt{ab}$. □

PROBLEM 32 Prove that for real a, b ,

$$a^4 + b^4 \geq 2a^2b^2.$$

Hint: AM-GM

Proof. By AM-GM,

$$a^4 + b^4 \geq 2\sqrt{a^4b^4} = 2a^2b^2.$$

□

PROBLEM 33 Prove that if $a, b, c > 0$ are real numbers, then

$$(a+b)(b+c)(c+a) \geq 8abc.$$

Hint: AM-GM.

Proof. By AM-GM, $a+b \geq 2\sqrt{ab}$, $b+c \geq 2\sqrt{bc}$, and $c+a \geq 2\sqrt{ca}$. Multiplying gives

$$(a+b)(b+c)(c+a) \geq 8\sqrt{a^2b^2c^2} = 8abc.$$

□

PROBLEM 34 (*) Prove that if $a, b, c > 0$ are real numbers, then

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3.$$

Hint: AM-GM.

Proof. By AM-GM, $\frac{a}{b} + \frac{b}{a} \geq 2$, and similarly for cyclic pairs. Summing the three inequalities gives

$$2 \left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \right) \geq 6,$$

so $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$. □

PROBLEM 35 (*) Prove that if $a, b, c > 0$ are real numbers, then

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9.$$

Hint: AM-GM

Proof.

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = 3 + \left(\frac{a}{b} + \frac{b}{a} \right) + \left(\frac{b}{c} + \frac{c}{b} \right) + \left(\frac{c}{a} + \frac{a}{c} \right).$$

By AM-GM, each pair is at least 2, so the total is at least $3 + 2 + 2 + 2 = 9$. □

PROBLEM 36 (*) Prove that if $a, b, c > 0$ are real numbers, then

$$\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+a} \geq 1.$$

Hint: Do not use AM-GM.

Proof. Since $a + b \leq a + b + c$, we have $\frac{a}{a+b} \geq \frac{a}{a+b+c}$, and similarly for the other terms. Therefore

$$\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+a} \geq \frac{a+b+c}{a+b+c} = 1.$$

We do not need AM-GM. □

PROBLEM 37 (**) Given $a + b = 1$, prove

$$ab \leq \frac{1}{4} \text{ and } a^2 + b^2 \geq \frac{1}{2}.$$

Hint: Use AM-GM on a^2 and b^2 .

Proof. By AM-GM on a^2 and b^2 ,

$$\frac{a^2 + b^2}{2} \geq \sqrt{a^2 b^2} = |ab|,$$

so $a^2 + b^2 \geq 2|ab|$. Since $a + b = 1$,

$$1 = (a + b)^2 = a^2 + b^2 + 2ab \geq 2|ab| + 2ab.$$

If $ab \geq 0$, then $1 \geq 4ab$, so $ab \leq \frac{1}{4}$; if $ab \leq 0$, then $ab \leq \frac{1}{4}$ holds. Then

$$a^2 + b^2 = 1 - 2ab \geq 1 - 2 \cdot \frac{1}{4} = \frac{1}{2}.$$

□

PROBLEM 38 (**) Given positive real numbers a, b, c with $abc = 1$, prove

$$a + b + c \geq 3.$$

Hint / Challenge: Use only 2-variable AM-GM.

Proof. Let $w = (abc)^{1/3} = 1$. By the 2-variable AM-GM,

$$\frac{a + b}{2} \geq \sqrt{ab}, \quad \frac{c + w}{2} \geq \sqrt{cw}.$$

Applying AM-GM again to these two averages gives

$$\frac{a + b + c + w}{4} \geq \sqrt{\sqrt{ab}\sqrt{cw}} = (abcw)^{1/4} = 1.$$

Thus $a + b + c \geq 3$.

□

PROBLEM 39 (****) Prove that for positive real numbers $x_1, \dots, x_n$,

$$\frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}.$$

Hint: Do Problems 4 and 21 first. Can you combine the two proofs?

Proof. First prove the case $n = 2^k$ by induction on k using only the 2-variable AM-GM. The case $k = 1$ is $n = 2$. If the statement holds for 2^k , apply it to the pairs

$$\frac{x_1 + x_2}{2}, \frac{x_3 + x_4}{2}, \dots, \frac{x_{2^{k+1}-1} + x_{2^{k+1}}}{2},$$

to get

$$\frac{x_1 + \dots + x_{2^{k+1}}}{2^{k+1}} \geq \sqrt[2^{k+1}]{x_1 x_2 \dots x_{2^{k+1}}}.$$

Now let n be arbitrary. Let $m = 2^k \geq n$ and set $y_i = x_i$ for $1 \leq i \leq n$ and $y_i = t$ for $n < i \leq m$, where $t = \sqrt[n]{x_1 \dots x_n}$. Then $\sqrt[m]{y_1 \dots y_m} = t$. Applying the m -variable case gives

$$\frac{x_1 + \dots + x_n + (m - n)t}{m} \geq t,$$

so $\frac{x_1 + \dots + x_n}{n} \geq t$.

□

PROBLEM 40 (**) Prove that for positive real a, b ,

$$\frac{a^3}{b^2} + \frac{b^3}{a^2} \geq a + b.$$

Hint: Apply AM-GM on three numbers.

Proof. Apply AM-GM to the three positive numbers $\frac{a^3}{b^2}, b, b$:

$$\frac{\frac{a^3}{b^2} + b + b}{3} \geq \sqrt[3]{\frac{a^3}{b^2} \cdot b \cdot b} = a,$$

so $\frac{a^3}{b^2} + 2b \geq 3a$. Similarly,

$$\frac{\frac{b^3}{a^2} + a + a}{3} \geq \sqrt[3]{\frac{b^3}{a^2} \cdot a \cdot a} = b,$$

so $\frac{b^3}{a^2} + 2a \geq 3b$. Adding gives

$$\frac{a^3}{b^2} + \frac{b^3}{a^2} \geq a + b,$$

the desired inequality. □